SQL MASTERY NOTEBOOK / DAY 01

Data & SQL Foundations

Start from zero | SQL Server / T-SQL | Venkata Siva

Your Day 1 learning path

Today's goal: Understand the data, understand the database, and run a SELECT you can explain. We use one online store throughout.

Step	Main question
1. Data	What facts do we record, and what do they mean?
2. Databases	Why keep related facts in a managed system?
3. Relational structure	How do tables, types and relationships organize them?
4. SQL and tools	How do people and applications request data?
5. Query journey	What does the database engine do with the request?
6. First SELECT	Can I predict, run and explain the result?

CONCEPT · The reading pattern

Each lesson gives you a plain definition, a concrete example and a memory cue. Technical words appear next to their meaning. Learn the idea first; use the word once you can explain it.

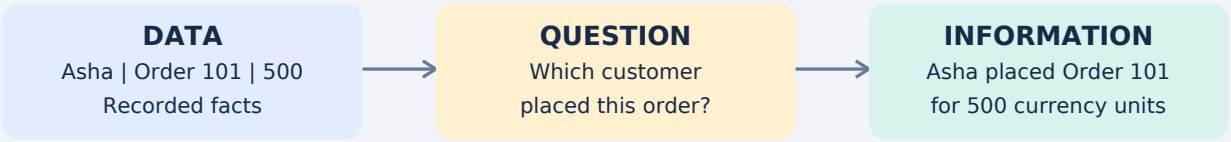
Read in blocks: Begin with the data lessons. Take a break before databases. Finish with the SQL lab, interview explanations and independent practice. Allow roughly 3–4 hours with practice, or split the reading across sessions.

Today's SQL focus: basic SELECT. Relationships and transactions are introductions; learn their purpose first. Day 2 adds filtering through IS NULL and IS NOT NULL.

Lab: The examples use Microsoft SQL Server / T-SQL. You need an existing SQL Server connection to execute them. The HTML activities are local teaching simulations.

02 Data becomes useful when it has meaning

Data is recorded facts: a name, a price, a date, a quantity. **Information** is the meaning we obtain when those facts answer a question.



DATABASE: organized collection of related data.

DBMS: software that stores, retrieves and manages that data.

SQL: language used to ask for data or define changes in a relational system.

Recorded fact	Context	What we learn
500	Order 101 total, in INR	Order 101 costs INR 500
Hyderabad	Asha’s current city	Asha’s profile lists Hyderabad
2	Quantity on an order line	Two units were ordered

The same number can mean different things. “500” might be an amount, a quantity or an identifier. The field name, units and business definition tell us how to interpret it.

Our store records customers, products and orders. A **database** is an organized collection of related data and objects used to manage those facts.

CONCEPT · Memory cue

Data is the recorded fact. Context tells you what it means. Information answers a question.

Interview phrase: “Data becomes meaningful when I know its context, such as units, time and what the record represents.”

03 Three forms of data, with everyday examples

These forms describe **how the main content is organized**. They do not describe whether a value is text or a number.

STRUCTURED
Defined record fields
Customer rows
Bank transactions

SEMI-STRUCTURED
Keys and nested values
JSON order response
XML invoice

UNSTRUCTURED
Free-form main content
Review text / delivery photo
Support-call audio

One business produces all three forms; classify the content, not the entire company.

Free-form content can still have structured metadata such as a rating or customer ID.

Form	Easy explanation	More examples
Structured	Records follow defined fields	Bank transaction table; patient appointments; product price list
Semi-structured	Keys or tags describe values; records can be nested or flexible	JSON order response; XML invoice; JSON delivery event
Unstructured	Main content is free-form rather than our predefined record fields	Customer review; scanned prescription image; support-call audio

Structured: predefined record fields

A customer table gives every row named columns: CustomerID, CustomerName and City. A consistently defined CSV or spreadsheet can also contain structured data.

Semi-structured: named keys and nesting

```
{"customerId": 1, "name": "Asha",
 "preferences": {"language": "English"}}
```

JSON is self-describing through keys and nesting. Different records can carry different fields; a schema can still validate them.

Unstructured: free-form main content

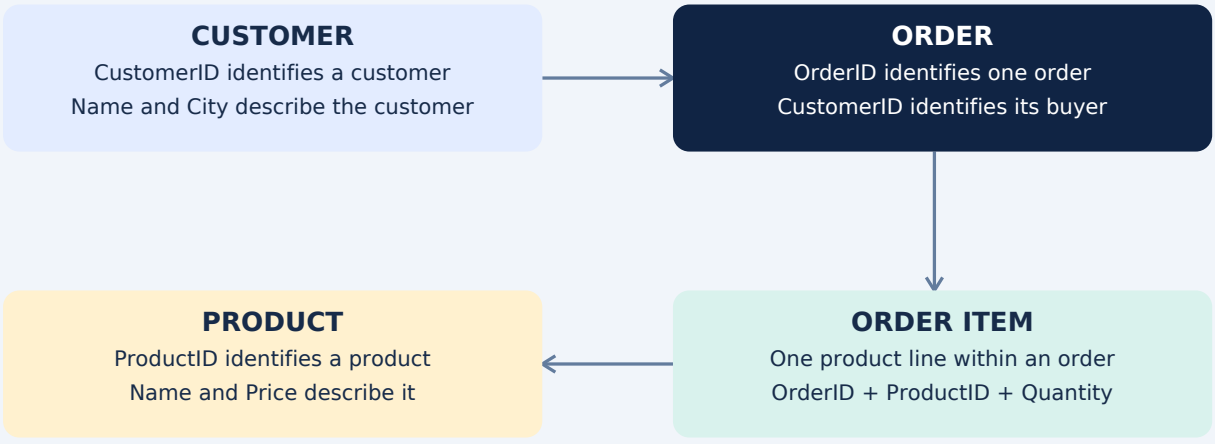
“Good product, but delivery was late” is review text. Its rating and customer ID can be structured fields alongside that text. An image also has a file format and metadata; unstructured does not mean completely without structure.

Memory cue: structured = fixed fields; semi-structured = keys or tags; unstructured = free-form main content.

04 Entities, attributes and identifiers

Before thinking about SQL syntax, identify the things your business records.

Term	Plain meaning	Store example
Entity	A thing or event we need to describe	Customer, product, order
Attribute	A property of that entity	Customer name or city
Identifier	A value that distinguishes one instance	CustomerID 1



Start with what one record represents. Detailed key and relationship design comes later.

Asha is one customer. CustomerName and City describe her. CustomerID distinguishes her from another person who might also be called Asha.

An order is a different entity. It needs its own identifier and facts such as its date and customer reference. An order item represents a product line within that order.

CONCEPT · Memory cue

Entity: what is it? Attribute: what describes it? Identifier: which one is it?

Do not use a name as a unique identifier just because it looks unique in a small sample. Names can repeat and can change.

05

Grain: say what one row represents

Grain means the level of detail represented by one row. State it as a sentence before counting or adding anything.

ONE ROW PER ORDER

OrderID | OrderTotal

101 | 500

Order 101 is counted once

ONE ROW PER ORDER ITEM

OrderID | Product | LineAmount

101 | P1 | 300

101 | P2 | 200

Correct line-level total for Order 101: 300 + 200 = 500.

If the whole order total 500 is copied to both lines and summed: 500 + 500 = 1,000.

The arithmetic is correct, but the meaning is wrong because the grain was mixed.

Order 101 contains a book costing 300 and a pen costing 200.

Dataset	One row represents	Rows for this example
Orders	One order	1 row, total 500
OrderItems	One product line in an order	2 rows, amounts 300 and 200

If you copy the whole-order total 500 onto both item rows and add it, you get 1000. The arithmetic works; the business answer is wrong. Add the line amounts instead: 300 + 200 = 500.

CONCEPT · Memory cue

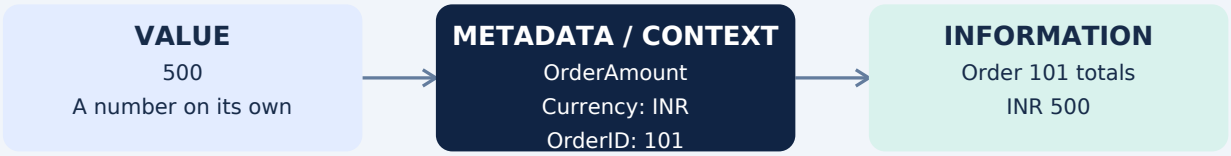
One row, one meaning. Counting rows only makes sense after you know that meaning.

Interview phrase: “I confirm the grain before counting, aggregating or joining data to avoid double counting.” You will learn aggregation and joins later.

06

Metadata tells you how to read a value

Metadata is information that describes data. It tells you what a field means and how to use it.



Technical metadata: column names, types and object definitions.

Business metadata: meaning, units, owner and accepted rules.

Operational metadata: source, collection time and update information.

Description	What it tells you
Column name: OrderAmount	What this number represents
Currency: INR	Which unit the amount uses
Definition: total for one order	The amount’s business meaning and grain
Source: checkout system	Where the value came from
Updated at: stated time	How fresh the value is

A **data dictionary** records agreed field definitions. It can document whether an amount includes tax or how a status is interpreted.

Time context also matters. Asha may live in Chennai now while an earlier order was delivered to Hyderabad. A current profile and a historical delivery record answer different questions; neither is automatically wrong.

Memory cue: a number needs a name, a unit and a purpose.

Interview phrase: “Metadata helps me interpret a dataset correctly rather than guessing from its values.”

07 Good structure does not guarantee good data

Data quality means whether the recorded data is suitable for the question you want to answer.

FORM / RULE CHECKS

City = Chennai
Known city spelling
Allowed text length
Passes basic validation

REAL-WORLD VERIFICATION

Customer confirms Hyderabad
Recorded city is wrong
A permitted value is not
proof of factual accuracy

Integrity rules protect what was defined. Quality also asks whether facts are fit for use.

Completeness, validity, consistency, uniqueness and timeliness each ask different questions.

Quality check	Question	Example
Accuracy	Is the fact correct?	Does the address match the customer’s address?
Completeness	Is a required fact present?	Is the delivery address recorded?
Validity	Does it follow the stated rule?	Is a quantity within the allowed range?
Consistency	Do related records agree?	Do shipping and billing records agree where required?
Uniqueness	Are unintended duplicates present?	Was one customer created twice?
Timeliness	Is it fresh enough?	Is stock information current enough for checkout?

Every address field may be filled while the address itself is wrong. That data is complete but inaccurate.

Integrity rules enforce defined conditions in the database. They cannot prove every real-world fact. For example, a valid city name can still be wrong for a particular customer.

Memory cue: present does not mean correct; correct format does not mean correct fact.

08 NULL, zero and empty text are different

NULL marks a value that has not been supplied or is not known. It is not the number zero and not empty text.

Recorded value	What it means in this example
Amount: 0	The recorded amount is known to be zero
Amount: NULL	No amount is recorded; do not assume zero
City: ''	Recorded text with zero characters
City: NULL	No city is recorded
City: 'NULL'	Literal text containing the letters N U L L

Meera’s City is NULL in our practice table. It does not mean she lives in a place named NULL or that her city is an empty string.

A missing value can be unknown, not yet collected or not applicable, depending on the field’s definition. Its exact meaning comes from context.

WARNING · Keep missingness meaningful

Do not replace missing values with zero or empty text simply to make a result look tidy. That can change what the data says.

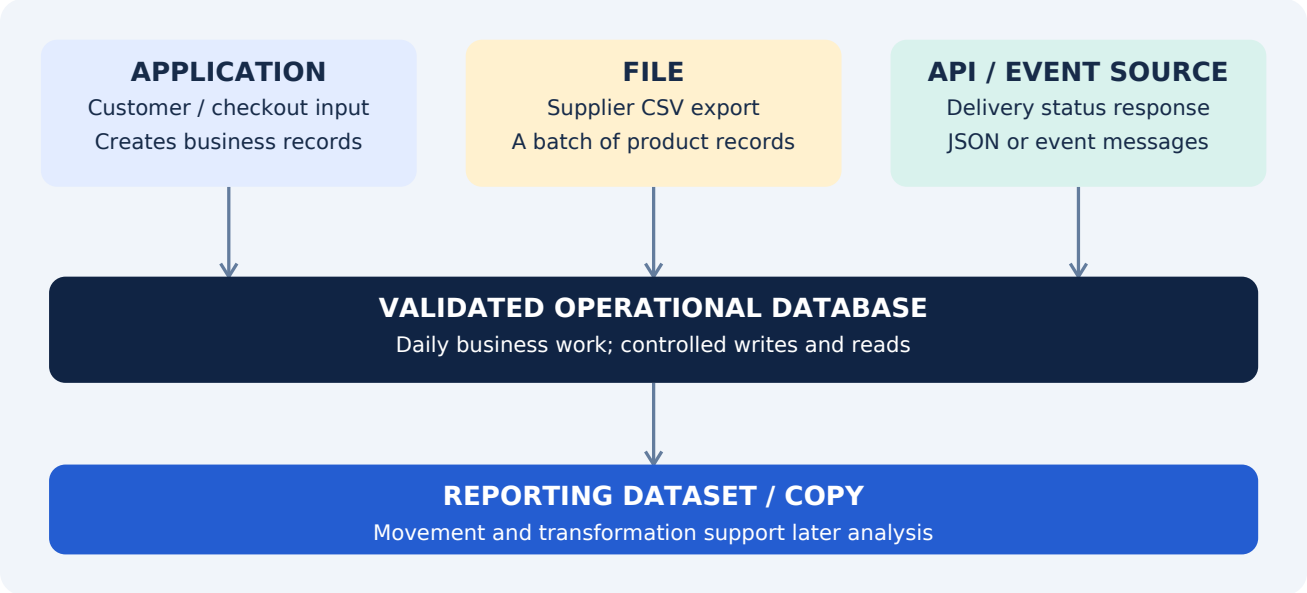
Memory cue: NULL = no known value; zero = a known number; empty = recorded text with no characters.

Day 2 teaches IS NULL and IS NOT NULL. Today, recognize and explain the difference.

09

Where data comes from and where it goes

Our store collects data from several places: a checkout form, a supplier CSV and a delivery API. An **API** is an interface through which programs exchange requests and responses.



Step	Store example
Capture	A customer submits an order
Validate	Check required fields and allowed values
Store	Save the accepted order
Use and update	Show its status and record shipment
Keep or remove	Preserve required history; follow retention rules

A report might use a copy refreshed last night. That copy cannot automatically claim to show current stock now. Know the source and refresh time.

A **backup** supports recovery. An **archive** preserves selected history. A **reporting copy** supports analysis. These purposes differ even though each may contain copied data.

Access habit: grant only the data access needed for the task. Exported files also need protection; database permissions do not automatically protect a new CSV.

Memory cue: source, meaning and freshness travel with the data.

10 Why files alone can become difficult

Files and spreadsheets are useful. The problem grows when many people or programs must maintain shared facts consistently.



Suppose sales, shipping and billing each keep Asha’s city. Shipping updates its copy, but the others do not. Now which city should the store use?

Problem	Why a managed database helps
Repeated facts	Share a controlled source for common facts
Conflicting edits	Coordinate concurrent changes
Different checks in each program	Enforce shared database rules
Difficult recovery	Provide transaction and recovery mechanisms
New questions need custom file logic	Use flexible queries over managed data

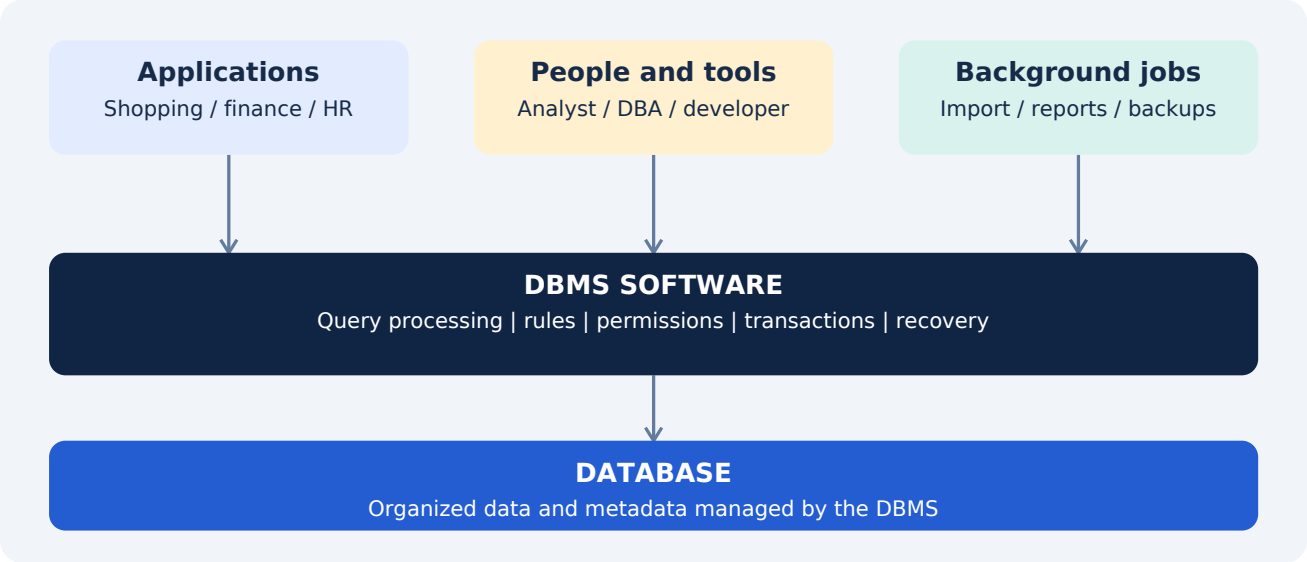
A DBMS brings these responsibilities into a shared system. Good design and administration are still needed; installing database software does not automatically make data accurate.

Memory cue: a shared fact needs shared rules and controlled changes.

A small personal list may work perfectly well as a file. Choose a database when the coordination, querying and reliability needs justify it.

11 Database, DBMS, RDBMS and SQL

These words describe different roles. Keep this distinction clear from your first day.



Word	Plain meaning	Example
Database	Organized data and database objects	The store’s customers and orders
DBMS	Software that manages databases	Software that controls reads, changes and access
RDBMS	A DBMS based on the relational model	SQL Server, PostgreSQL or MySQL
SQL	Language used to express database operations	SELECT CustomerName ...
T-SQL	SQL Server’s SQL dialect and extensions	The syntax used in this handbook

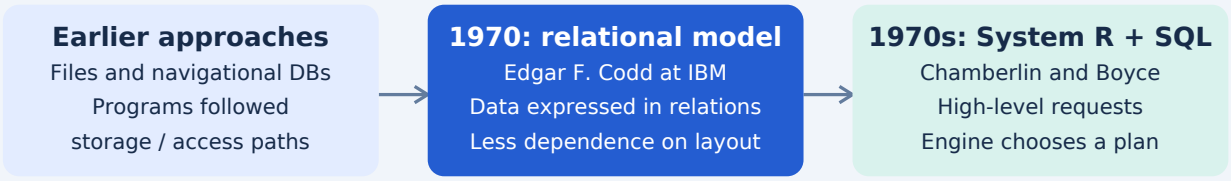
An RDBMS is a kind of DBMS. It represents data using relational structures, commonly presented as tables, and supports operations and rules over that data.

> **WARNING · Avoid the common shortcut**
>
> DBMS does not mean “single user,” and RDBMS does not simply mean “multiple users.” The relational model is the key distinction.

Memory cue: database stores; DBMS manages; RDBMS is relational; SQL expresses the request.

12 Why SQL and relational databases emerged

Earlier systems often required programmers to understand storage layouts or navigate particular record paths. Asking a new question could require more custom access code.



Then: describe how to navigate data.

Relational direction: describe the data and relationships.

SQL direction: state the result you want; the engine plans the work.

Files and nonrelational systems still have useful roles; this is not a replacement timeline.

1970: Edgar F. Codd published the relational model while working at IBM. Its central direction was to describe data through relations and reduce dependence on physical storage organization.

1970s: IBM's System R project explored a working relational system. Donald Chamberlin and Raymond Boyce developed the language that became SQL, initially called SEQUEL.

Why it matters now: you can ask for customer names without coding the physical steps for finding every stored record. The engine plans the access work.

This idea is related to **data independence**: separating the logical view of data from its physical storage details. It does not mean every change to a table definition leaves applications unaffected.

Memory cue: ask for the result; avoid hard-coding every storage access step.

Files and nonrelational systems remain useful. This is a conceptual history, not a claim that one technology replaced every other one.

13 Read a table across and down

A **table** organizes records into named columns and rows. Our sample table has one row per customer.

CustomerID	CustomerName	City
1	Asha	Hyderabad
2	Ravi	Chennai
3	Neha	Hyderabad
4	Omar	Mumbai
5	Meera	NULL

Across row 2: Ravi is CustomerID 2 and his recorded city is Chennai. **Down City:** each row supplies the city value for a different customer.

Word	Meaning
Row / record	One instance at the table’s grain
Column / attribute	One named property
Table definition	Column names, types and configured rules
Result set	Rows and columns returned by a query

Hyderabad repeats because two customers share the city. A column is not automatically unique.

Order matters in different ways: a SELECT list determines output-column order. A table’s stored rows do not come with a guaranteed query display order; requesting row order is a later lesson.

Memory cue: across = one record; down = one attribute.

14 Data types: represent the right kind of value

A **data type** describes how a value is represented and which values that representation allows. Choose the type from the meaning of the field.

SQL Server type	Useful for	Example
int	Whole numbers	Quantity 2 or CustomerID 1
decimal(10,2)	Exact decimal amounts	500.25
date	Calendar date	2026-10-13
varchar(30)	Variable-length character data	Hyderabad
nvarchar(30)	Unicode character data	Names requiring Unicode representation

For decimal(10,2), precision 10 is the total allowed digits; scale 2 is the digits after the decimal point. You do not need to memorize every type or range today.

A code such as '00125' may need text to preserve its leading zeros. An identifier made of digits is not automatically a quantity you should add.

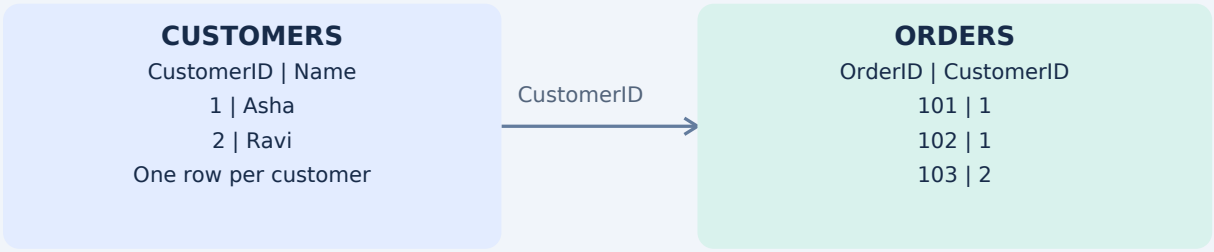
The type cannot explain everything: an amount also needs a currency and business definition. Nor can a text type prove that the recorded city is correct.

Memory cue: meaning first, type second.

Detailed encoding and length rules are later topics. Our lab uses int and varchar for simple English sample values.

15 Relationships connect facts; constraints protect rules

A customer can place several orders. Keep the shared customer profile in Customers, and let Orders refer to its CustomerID.



Customer 1 has Orders 101 and 102; Customer 2 has Order 103.

Keep customer details once, then reference the customer from each order.

A foreign-key rule can reject an order that refers to a nonexistent customer.

Asha’s two orders can refer to CustomerID 1. We can update her current profile once rather than maintain several independent copies. A historical delivery address may still be deliberately saved on an order.

Constraint	Rule it can enforce
Primary key	Identify each row uniquely; required key values
Foreign key	Validate a reference to a permitted key
NOT NULL	Require a value
CHECK	Reject values that violate a defined condition
UNIQUE	Protect a defined uniqueness rule

Referential integrity means keeping defined references valid. For example, an order’s required customer reference should not point to a nonexistent customer.

WARNING · Design the rules deliberately

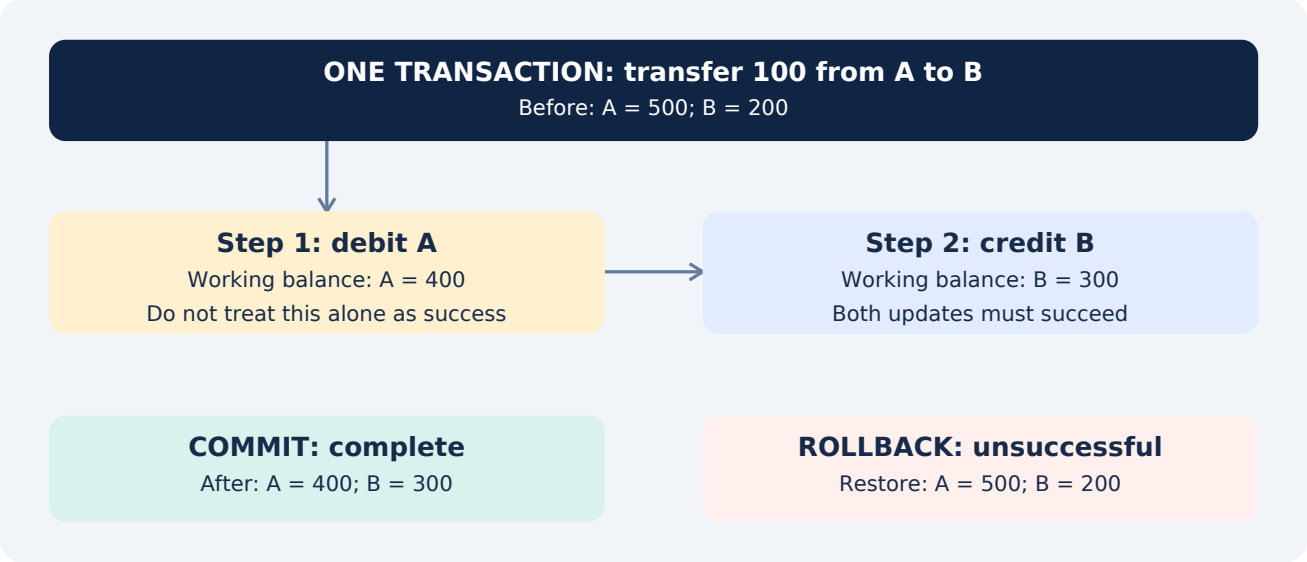
Matching column names do not create a relationship or a constraint. The designer defines which fields refer to which keys and what rules to enforce.

Memory cue: relationships connect; constraints enforce. Detailed key design and joins come later.

16 Transactions and OLTP: daily business work

OLTP: Online Transaction Processing supports daily operations such as orders, bookings, payments and stock updates. Requests are often short and frequent. “Online” does not require a public website.

A **transaction** coordinates database operations as one unit of work. Transferring 100 from account A to B needs both the debit and the credit.



ACID property	Plain meaning
Atomicity	Required operations succeed together or do not remain partly completed
Consistency	The completed transaction preserves defined rules
Isolation	Concurrent transactions interact under the chosen isolation rules
Durability	Normally committed, durable changes survive supported recovery

Isolation coordinates simultaneous work; it does not mean all users must take turns. Its exact behavior depends on the isolation settings.

A SELECT can be an OLTP read, such as retrieving one order’s status. OLTP describes a workload, not one SQL command.

Memory cue: one business action can need several database operations.

17 OLTP and analytics answer different questions

OLAP: Online Analytical Processing focuses on analysis, trends and summaries across data. Compare the questions before comparing technologies.

	Operational / OLTP style	Analytical / OLAP style
Question	What is Order 101’s status?	Which products sold most this year?
Work	Short reads and changes	Larger reads and summaries
Purpose	Run the business	Understand the business
Typical users	Applications and operational staff	Analysts and reporting tools

An RDBMS can support both kinds of work. Large reports can compete with operational requests, so some systems use a separate reporting dataset. Such a copy may have a refresh delay.

Relational-system strengths: flexible querying, shared rules, controlled access and transaction support.

Tradeoffs: table design needs care; structure changes need planning; expensive queries consume resources; backup and maintenance are necessary. No product choice removes these responsibilities automatically.

Memory cue: OLTP handles the activity; analytics studies the activity.

Today, recognize the difference. Warehouse design and performance tuning belong to later lessons.

18 SQL describes the work you want

SQL: Structured Query Language expresses operations involving data and database objects. SQL Server accepts **T-SQL: Transact-SQL**, its SQL dialect and extensions.

“Show the customer names” becomes:

```
SELECT CustomerName
FROM #Customers;
```

You will run this example after the supplied lab setup.

This is **declarative**: you describe the requested result rather than every physical retrieval step. The engine decides how to perform the access work.

Need	Commands to recognize
Read rows	SELECT
Add or change rows	INSERT, UPDATE, DELETE
Define or change objects	CREATE, ALTER, DROP
Manage access	GRANT, DENY, REVOKE in T-SQL
Coordinate transactions	BEGIN TRANSACTION, COMMIT, ROLLBACK

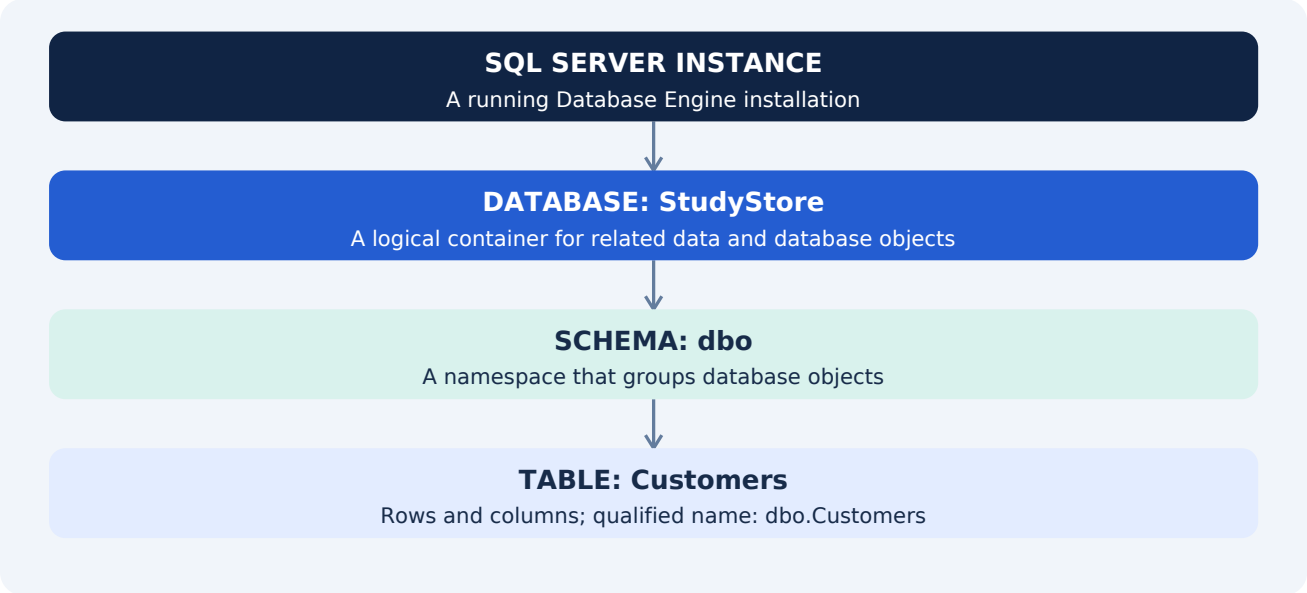
Interview labels include DDL for definition, DML for data manipulation, DCL for access and TCL for transactions. SELECT is sometimes called DQL and sometimes classified within DML. Explain the purpose rather than treating the labels as universal rules.

Memory cue: SQL says what; the engine plans how.

Valid SQL can still answer the wrong business question or run inefficiently. SQL Server, PostgreSQL and MySQL also differ in syntax and features.

19 SQL Server is the engine; SSMS is the client

SQL Server manages databases and executes requests. **SSMS: SQL Server Management Studio** is a client tool for writing SQL, connecting to the engine and displaying results.



Name	What it means
SQL Server instance	A separately configured Database Engine instance
StudyStore	Our conceptual database
dbo	A schema: a namespace for database objects
Customers	A table in that schema
dbo.Customers	Schema-qualified table name

“Schema” can also mean a data structure definition in broader discussions. In dbo.Customers, it means the object namespace.

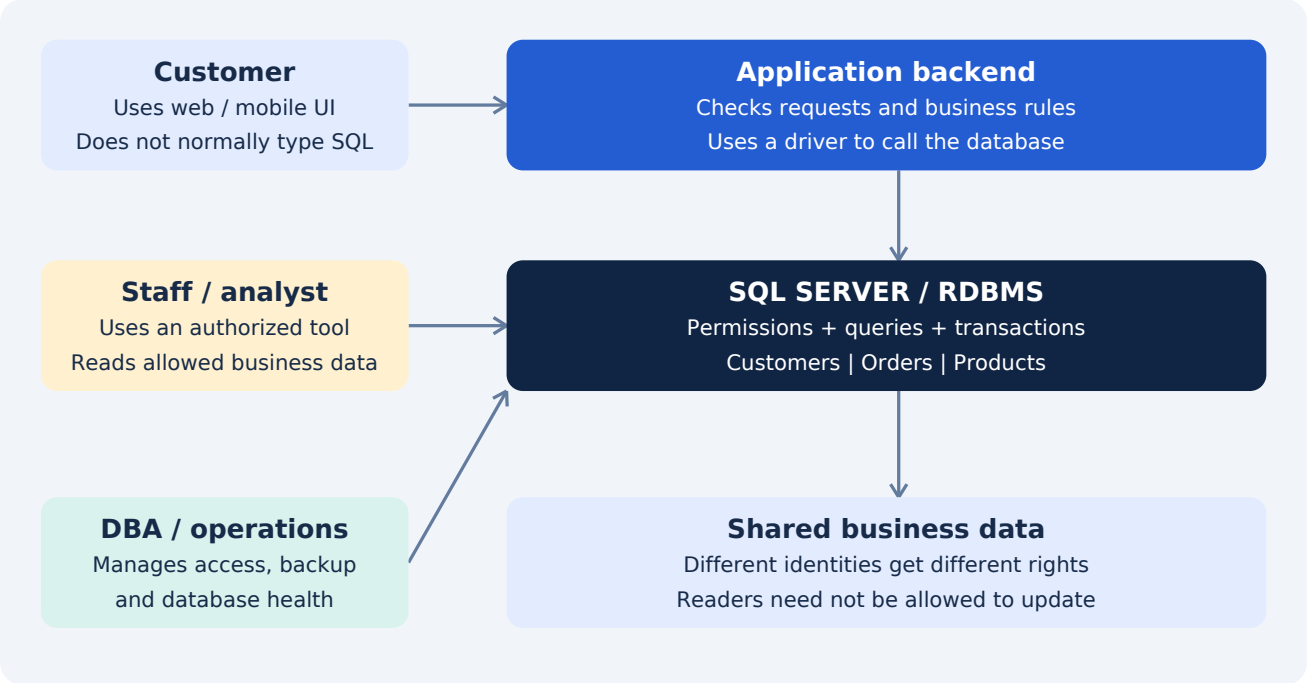
Opening SSMS does not create a server. Closing SSMS does not normally stop SQL Server; other clients can continue using it.

Our lab uses `#Customers`, a **local temporary table** held in tempdb and scoped to the connection. It is a stand-in for a permanent business table; use the same live connection for setup and SELECT.

Memory cue: SSMS sends; SQL Server serves.

20 How applications and people use the database

A shopper uses a screen; the application backend sends the database request. A **backend** is the application logic serving that screen. A **driver** or library handles database communication.



Follow "My orders": the backend identifies the customer, checks which orders they may see, asks the database, and formats the returned rows for the screen.

Party	Typical role
Customer	Uses the application
Developer	Builds application logic and database requests
Analyst	Reads authorized data for analysis
DBA	Manages database access, backup, recovery and health
Automated job	Runs scheduled imports or reporting

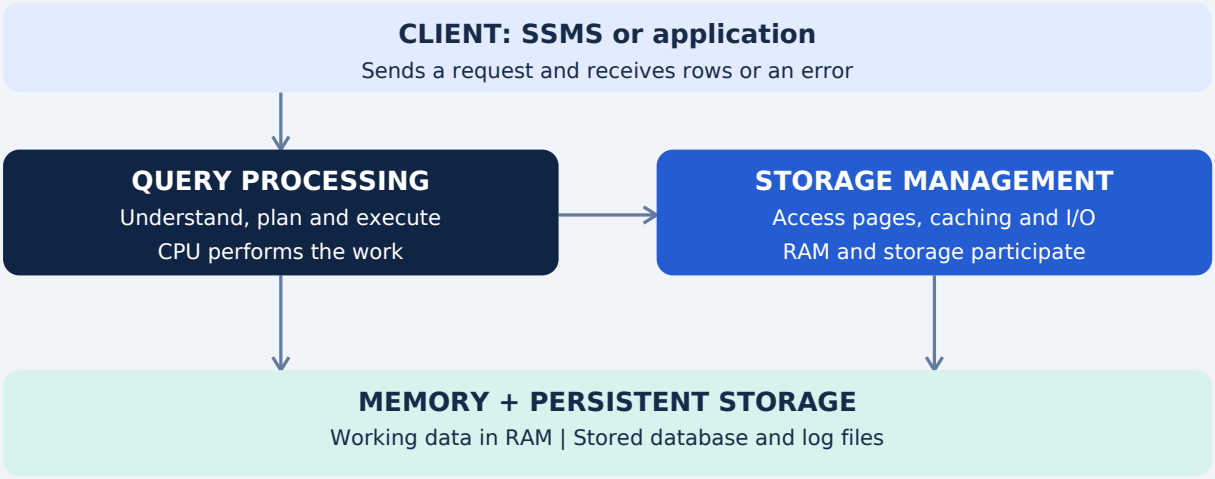
Authentication establishes identity. **Authorization** decides what that identity may do. Application rules and database permissions both matter.

Applications should pass user-supplied values as parameters rather than concatenate them into raw SQL text. Detailed parameterized queries come later.

Memory cue: screen → backend request → database result → screen. This is a common pattern, not the only possible architecture.

21 The engine uses memory and persistent storage

A table is a logical object. Its data is managed in database structures; it is not a CSV that every SELECT reads line by line.



Tables are logical objects. Physical storage uses structures such as files and pages.

Component	Its role
Query processor	Understands the request and runs a plan
Storage management	Accesses the needed database data
Buffer cache	Memory holding data pages
Persistent files	Stored database data and transaction logs
Transaction log	Change records used for transaction handling and recovery

A **data page** is a unit of stored database data. Day 1 does not require page sizes or file-layout details.

The needed page may already be in memory, so a SELECT does not always require a fresh disk read. A usable execution plan may also be cached.

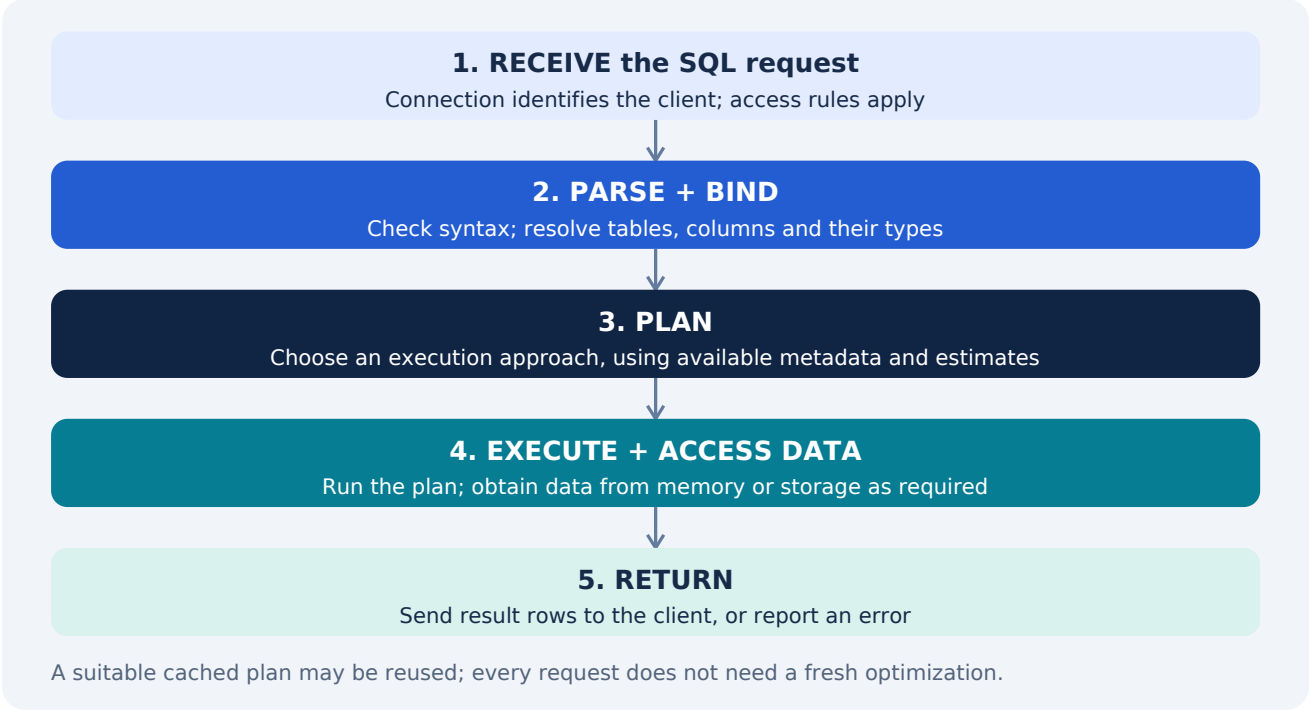
WARNING · Plan cache is different from result caching

A cached plan describes how to do the work. It does not automatically contain the old rows to return again.

Memory cue: memory holds working data; storage preserves files; the plan describes work.

22 What happens when SELECT executes?

Imagine asking for CustomerName from the sample table. Here is a simplified journey for a new query.



A suitable cached plan may be reused; every request does not need a fresh optimization.

Translate the technical words

Stage	Meaning	Example
Receive	Accept a client request	SSMS sends the SQL text
Parse	Check SQL syntax	Are the commas and clauses valid?
Bind	Resolve object and column names	Does CustomerName exist here?
Plan	Choose an execution strategy	Decide how to access the data
Execute	Perform the planned work	Obtain the requested values
Return	Send results to the client	Display names in the Results pane

The **query optimizer** considers ways to perform the request. An **execution plan** describes the chosen strategy. A suitable cached plan may be reused rather than planned afresh.

Access permissions also apply. A correctly written query can be denied, and a request can fail before producing a result.

Memory cue: parse syntax; bind names; plan work; execute; return. This is a teaching model, not a promise that every request follows identical internal steps.

23 Written SQL and its meaning are different

You write SELECT before FROM, but to understand this simple request, identify the source first and then the requested columns.

```
SELECT CustomerName
FROM #Customers;
```

LOGICAL MEANING

FROM: identify source rows
SELECT: choose output columns

Understand the requested result

PHYSICAL EXECUTION

Engine uses an execution plan
Access method depends on data and available structures

Operations may be rearranged

Writing SELECT first does not mean the engine physically performs it first.

No ORDER BY means no guaranteed row order, even if results look stable.

FROM #Customers: use these source rows. **SELECT CustomerName:** produce that output column.

A **physical execution plan** is how the engine actually performs the request. The SQL text does not prescribe every access step.

Three ideas stay separate: written clause order, logical meaning and physical execution. Optimizers can rearrange work while preserving the requested meaning.

Important: output-column order follows the SELECT list. Row order is not guaranteed without an explicit ordering request. We will learn ordering later.

Memory cue: SELECT says what to show; FROM says where the data comes from.

24 Prepare the practice table once

Open a fresh SQL Server query window, such as SSMS. Run this supplied setup once and keep that connection open.

```sql
CREATE TABLE #Customers (
    CustomerID   int NOT NULL PRIMARY KEY,
    CustomerName varchar(30) NOT NULL,
    City         varchar(30) NULL
);

INSERT INTO #Customers
    (CustomerID, CustomerName, City)
VALUES
    (1, 'Asha',  'Hyderabad'),
    (2, 'Ravi',  'Chennai'),
    (3, 'Neha',  'Hyderabad'),
    (4, 'Omar',  'Mumbai'),
    (5, 'Meera', NULL);
```

CREATE TABLE supplies the structure. INSERT supplies five sample rows. CustomerID must be present and unique; CustomerName is required; City may be NULL. You will learn to write these definitions later.

WARNING · Execute selected queries after setup

Do not rerun CREATE TABLE for every SELECT. If the temporary table already exists, start a fresh practice connection to restart. A different connection cannot see this session’s local table.

Verification scope: the SELECT results were checked with equivalent portable SQL in a local engine. The SQL Server temporary-table setup was not executed against a live SQL Server during preparation. Complete that step in your own SQL Server session.

25 Run your first SELECT and read the result

Ask for all three customer columns:

```
SELECT CustomerID, CustomerName, City
FROM #Customers;
```

SQL part	Meaning
SELECT	Start the output list
CustomerID, CustomerName, City	Return these columns in this order
Commas	Separate output items
FROM #Customers	Name the source
Semicolon	End the statement

Execute this query in your existing lab connection. In SSMS, use Execute or F5.

CustomerID	CustomerName	City
1	Asha	Hyderabad
2	Ravi	Chennai
3	Neha	Hyderabad
4	Omar	Mumbai
5	Meera	NULL

Expect five rows and three columns. The displayed teaching order is not a SQL row-order guarantee. Meera’s missing city remains NULL; the client may display it specially.

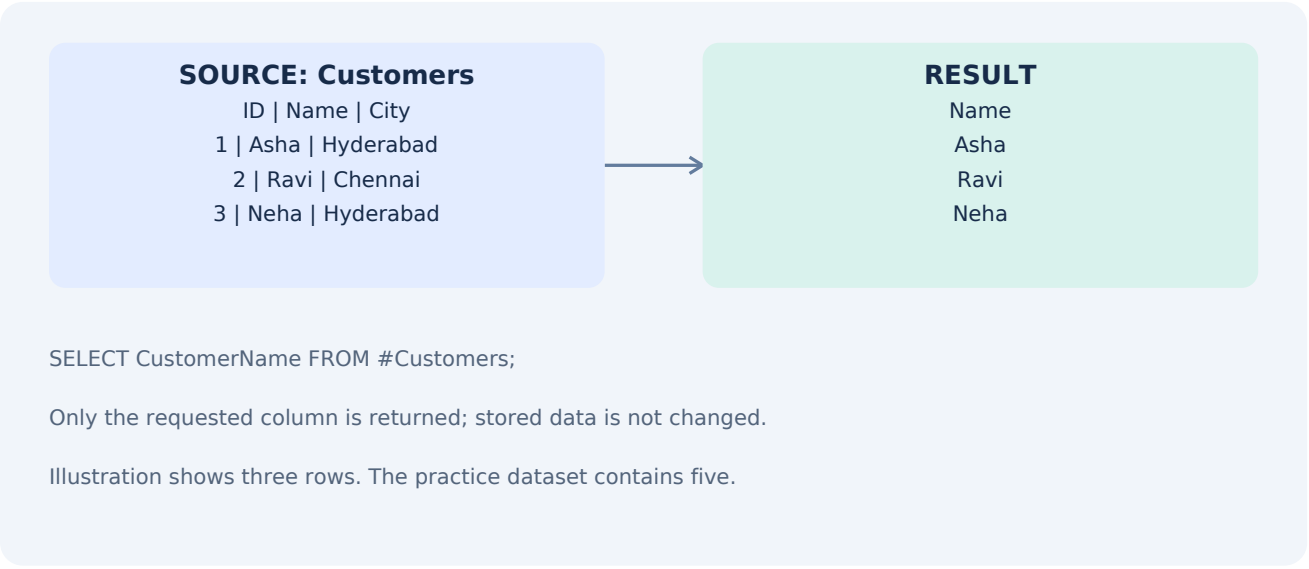
This plain SELECT reads the table; it does not change stored values.

Memory cue: predict the columns and row count before pressing Execute.

26 Choose columns without changing the table

Return only names:

```
SELECT CustomerName
FROM #Customers;
```



SELECT CustomerName FROM #Customers;

Only the requested column is returned; stored data is not changed.

Illustration shows three rows. The practice dataset contains five.

The result has one column and five rows. The source still has CustomerID, CustomerName and City. Choosing output columns is called **projection**.

What about SELECT *?

```
SELECT *
FROM #Customers;
```

The star returns the source’s columns. It is convenient for quick exploration. In maintained queries, name the columns you need so the intended output is clear.

Selecting City alone returns Hyderabad twice because two customers have that city. SELECT does not automatically remove duplicates; DISTINCT is a later topic.

Memory cue: fewer output columns does not mean fewer source rows or fewer stored columns.

27 First-day habits and common mistakes

Habit	Apply it to this lab
Know the grain	One row represents one customer
Choose the output	Name the columns you need
Predict the result	Five rows; requested columns only
Check the connection	Keep #Customers in its original session
Keep NULL meaningful	Meera’s city is missing, not empty text
Avoid row-order assumptions	Compare values, not screen positions
Read the actual error	Fix the reported issue before guessing

Symptom	First check
Invalid object name	Did setup run in this live connection?
Invalid column name	Is the column spelled correctly?
Incorrect syntax	Is a comma missing or a clause incomplete?
Table already exists	Did you run setup twice?
Unexpected output	Which SQL text was selected and executed?

Interview habit: explain what you requested, what you expected and why the result is correct. A successful query is not automatically a correct business answer.

28 The memory map: connect roles before terms

Memory cue	Reconstruct the meaning
Data plus context	Recorded facts become useful information
Fixed fields; keys; free form	The three forms of data
One row, one meaning	Grain
Database stores; DBMS manages	Collection versus software
Relationships connect; constraints enforce	Related facts and shared rules
SQL asks; engine plans	Declarative querying
SSMS sends; SQL Server serves	Client versus engine
NULL unknown; zero known	Different recorded states
Parse, bind, plan, execute, return	Simplified query journey
SELECT shows; FROM supplies	Output columns and source

Reconstruct the whole store example

The store records customers and orders. Tables give the records structure; relationships connect them; constraints protect defined rules. A client sends T-SQL to SQL Server. The engine processes the request, accesses data and returns a result set.

PRACTICE · Recall, then check

Close the notes and explain one cue with a customer example. Check your explanation, correct what you missed, then revisit it during your next study session.

These cues support understanding; they are not substitutes for executing and explaining queries.

29 Interview room: data and database foundations

Answer with a definition and one example.

ANSWER · 1. How do the three forms of data differ?

Structured data follows defined fields, like a customer table. Semi-structured data uses keys or tags, like JSON. Unstructured main content is free-form, like review text or audio. A review can also have structured metadata such as rating and CustomerID.

ANSWER · 2. Why does grain matter?

It tells me what one row represents. One order and one order item have different grains. Counting item rows does not automatically count orders, and summing a copied order total can double count.

ANSWER · 3. Database versus DBMS versus RDBMS?

A database holds organized data and objects. A DBMS manages databases. An RDBMS is a DBMS based on the relational model, commonly presented through tables. SQL Server is an RDBMS product.

ANSWER · 4. How do relationships help integrity?

Shared identifiers connect related records. A defined foreign key can enforce permitted references, such as an order referring to an existing customer. Equal column names alone do not enforce the rule.

ANSWER · 5. Can valid data still be inaccurate?

Yes. Hyderabad is a valid city value but can be wrong for a customer who lives in Chennai. Defined constraints protect rules; accuracy may require external verification.

30 Interview room: SQL and query behavior

Keep the explanation short and connect each term to its job.

ANSWER · 1. SQL, T-SQL, SQL Server and SSMS?

SQL is the language. T-SQL is SQL Server’s dialect and extensions. SQL Server is the database engine product. SSMS is a client tool used to send requests and inspect results.

ANSWER · 2. What does declarative mean?

I describe the requested result rather than every physical step for finding the data. For example, SELECT CustomerName asks for names; the engine plans the access work.

ANSWER · 3. What happens when SELECT runs?

For a simplified new-query path, the engine receives the request, checks syntax, resolves names, chooses a plan, executes it and returns results. A suitable existing plan can be reused. Identity and permissions also govern access.

ANSWER · 4. Is NULL the same as zero or empty text?

No. NULL marks no known or supplied value. Zero is a known number; empty text is a recorded string with zero characters. Their meanings should not be replaced interchangeably.

ANSWER · 5. Does SELECT guarantee order or remove duplicates?

No. Our simple SELECT returns the requested columns without changing the source. It does not promise row order or automatically remove duplicate values.

ANSWER · 6. What is OLTP?

A workload supporting daily operational events, such as purchases and bookings, with many short requests. A SELECT for an order status can be part of OLTP; OLTP is not a database product or SQL command.

31 Technical vocabulary in one glance

Word	What it means
Entity	The thing or event being described
Attribute	A property of an entity
Grain	What one record represents
Metadata	Descriptions that help interpret data
Constraint	A database-enforced rule
Referential integrity	Defined references stay valid
Transaction	Operations coordinated as a unit
Concurrency	Work overlaps in time
Schema	In SQL Server, an object namespace
Query	A request involving database data
Result set	Returned rows and columns
Parser	Checks and interprets SQL syntax
Binding	Resolves referenced objects and columns
Optimizer	Chooses an execution approach
Execution plan	Strategy for performing a request
Projection	Choosing output columns

Use the word only if you can give an example. “Projection” becomes clearer when you show SELECT CustomerName and explain that it returns one column while preserving the source table.

32 Sources and reading beyond Day 1

Primary references for the concepts and syntax. The visuals are original teaching illustrations, not actual captured execution plans.

Topic	Reference
Data and forms	IBM: data
Unstructured examples	IBM: examples and uses
Relational history	IBM: relational database
Relational model	IBM Research: Codd’s publication
SELECT syntax and logical meaning	Microsoft: SELECT
Plans and query processing	Microsoft: query architecture
Transactions and OLTP	Microsoft: OLTP
NULL	Microsoft: NULL and UNKNOWN
Supplied lab setup	Microsoft: CREATE TABLE
Log and recovery	Microsoft: transaction log

Later: filtering, sorting, joins, aggregation and detailed design. The remaining pages are independent practice, answers and your completion checklist.

33 Final practice: explain, predict and write

Try every question before opening the answer guide.

Data and database reasoning

1. Classify a customer table, a JSON order response, a delivery photo and a support-call recording. Explain why.

2. A review has text, rating 4 and CustomerID 1. Can its parts have different data forms?

3. Order 101 has two product lines, 300 and 200. How many order rows and item rows are needed? What is the correct total?

4. Every address field is filled. Does that prove accuracy?

5. Explain database, DBMS, RDBMS and SQL using one store example.

6. Why can an order reference CustomerID instead of copying an independent profile?

7. Explain what ACID protects in a transfer.

8. Distinguish a current order-status request from a yearly sales summary.

SQL and result reasoning

9. Explain SQL Server versus SSMS.

10. Trace a new SELECT from the client to the result.

11. Why can a cached plan still require data access?

12. Write a query returning CustomerID and CustomerName from #Customers.

13. Write a query returning CustomerName first and CustomerID second.

14. Return City alone. Predict rows, duplicates and NULL.

15. Explain whether those queries change the source table or guarantee row order.

Practice method: write your prediction, execute the query where applicable, then explain any disagreement.

34 Answer guide: data and database reasoning

ANSWER · 1–2. Data forms

A table is structured because it uses defined fields. A JSON response is semi-structured because keys and nesting describe its content. A photo and call recording have unstructured main content. Review text can be free-form while rating and CustomerID are structured metadata.

ANSWER · 3. Grain and total

One order row and two order-item rows. The line amounts sum to 300 + 200 = 500. Copying the whole-order total onto both lines and summing it would wrongly give 1000.

ANSWER · 4. Completeness and accuracy

Filled fields support completeness, not proof of correctness. An address can be fully populated but belong to the wrong location.

ANSWER · 5–6. Database roles and references

The database holds store facts. A DBMS manages them; an RDBMS manages relational structures. SQL expresses operations. Shared customer references reduce independently maintained profile copies; defined constraints enforce the references. Historical snapshots may still be stored deliberately.

ANSWER · 7–8. Transactions and workloads

Atomicity keeps both transfer steps together; consistency preserves defined rules; isolation coordinates concurrent interactions; durability preserves normally durable commits through supported recovery. Order status is an OLTP-style read. Yearly sales analysis is an analytical-style request. Either can use SQL.

35

Answer guide: your first SQL queries

9–11: SSMS is the client; SQL Server is the engine. A simplified new-query journey is receive, parse, bind, plan, execute and return. A plan describes work, not cached result rows, so data access remains necessary.

12. ID and name:

```
SELECT CustomerID, CustomerName
FROM #Customers;
```

13. Name first:

```
SELECT CustomerName, CustomerID
FROM #Customers;
```

Each returns five rows and two columns. The output columns follow the SELECT list.

14. City alone:

```
SELECT City
FROM #Customers;
```

Five rows, one column: Hyderabad appears twice; Chennai and Mumbai once each; one value is NULL. Display order is unspecified.

15: These plain SELECT queries read the source without changing its stored rows or columns. They neither guarantee row order nor automatically remove duplicates.

36 Your Day 1 completion checklist

Mark only what you can demonstrate.

[ ] I explain the three forms of data with different examples.

[ ] I identify an entity, its attributes and identifier.

[ ] I state what one row represents before reasoning about data.

[ ] I explain metadata, quality and missingness simply.

[ ] I distinguish database, DBMS, RDBMS and SQL.

[ ] I explain why relationships and defined constraints help.

[ ] I recognize OLTP, analytics and a transaction’s purpose.

[ ] I distinguish SQL Server, T-SQL and SSMS.

[ ] I trace the simplified query journey.

[ ] I run the supplied setup and SELECT queries in SQL Server.

[ ] I predict and explain output columns, rows, duplicates and NULL.

Final explanation: “I use a client to send T-SQL to SQL Server. The engine manages the database, processes the request, accesses data and returns a result set. My SELECT list determines the output columns.”

Day 2: selecting columns and filtering rows with comparisons, AND, OR, parentheses, IN, IS NULL and IS NOT NULL.

If you have only read the examples or used the HTML simulator, leave the real execution checkpoint incomplete.